
📖 Chapter 12 – CSS 3D Transforms: Zameen Aasmaan Ghuma Denge 🌐

❑ What is 3D Transform?

3D transform CSS me element ko **X, Y ke saath Z axis** pe bhi control karne deta hai.

Matlab:

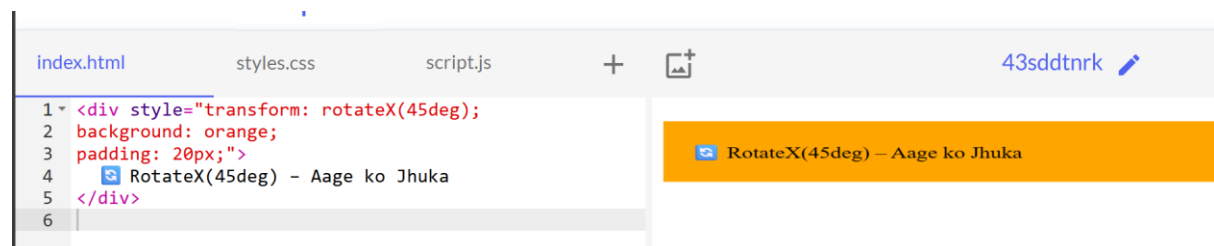
- **X** → left/right
- **Y** → up/down
- **Z** → andar/bahar (3rd dimension!) 😊

🌀 Basic 3D Properties:

Property	Kya karta hai
<code>rotateX(deg)</code>	X-axis ke around ghumata hai
<code>rotateY(deg)</code>	Y-axis ke around ghumata hai
<code>rotateZ(deg)</code>	Z-axis ke around (normal 2D rotate)
<code>perspective</code>	3D ka depth effect deta hai
<code>transform-style</code>	Child elements 3D me dikhaye ya flat

❑ Example 1: RotateX – Palti Maarna

```
<div style="transform: rotateX(45deg); background: orange; padding: 20px;">  
  🌀 RotateX(45deg) – Aage ko Jhuka  
</div>
```



❑ Example 2: RotateY – Side Se Dikhega

```
<div style="transform: rotateY(60deg); background: tomato; padding: 20px;">
  🔄 RotateY(60deg) – Side View Aaya
</div>
```



👁️ Perspective – 3D Depth Feel Deta Hai

perspective: parent pe lagate hain
Depth control karta hai – jaise camera distance

```
<div style="perspective: 800px;">
  <div style="transform: rotateY(60deg); background: purple; padding: 20px;
  color: white;">
    👁️ Yeh 3D depth ke saath ghooma
  </div>
</div>
```

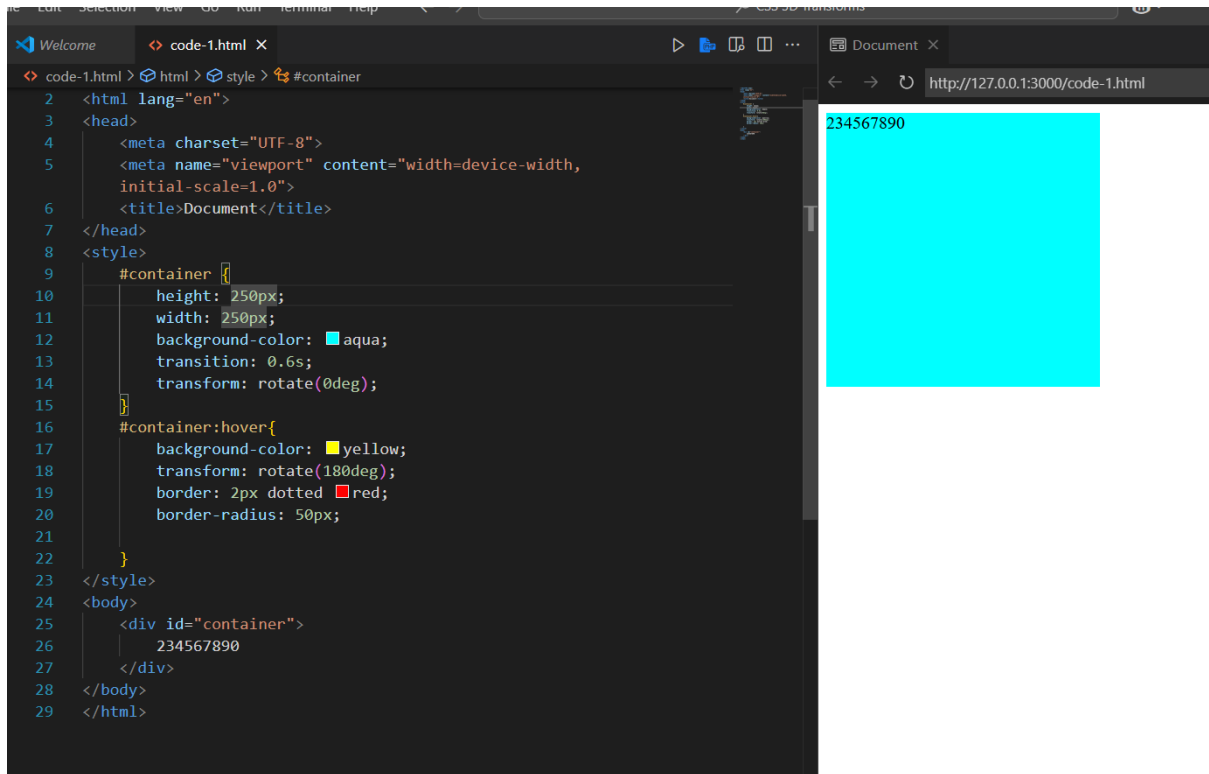


❑ Lower value = zyada depth

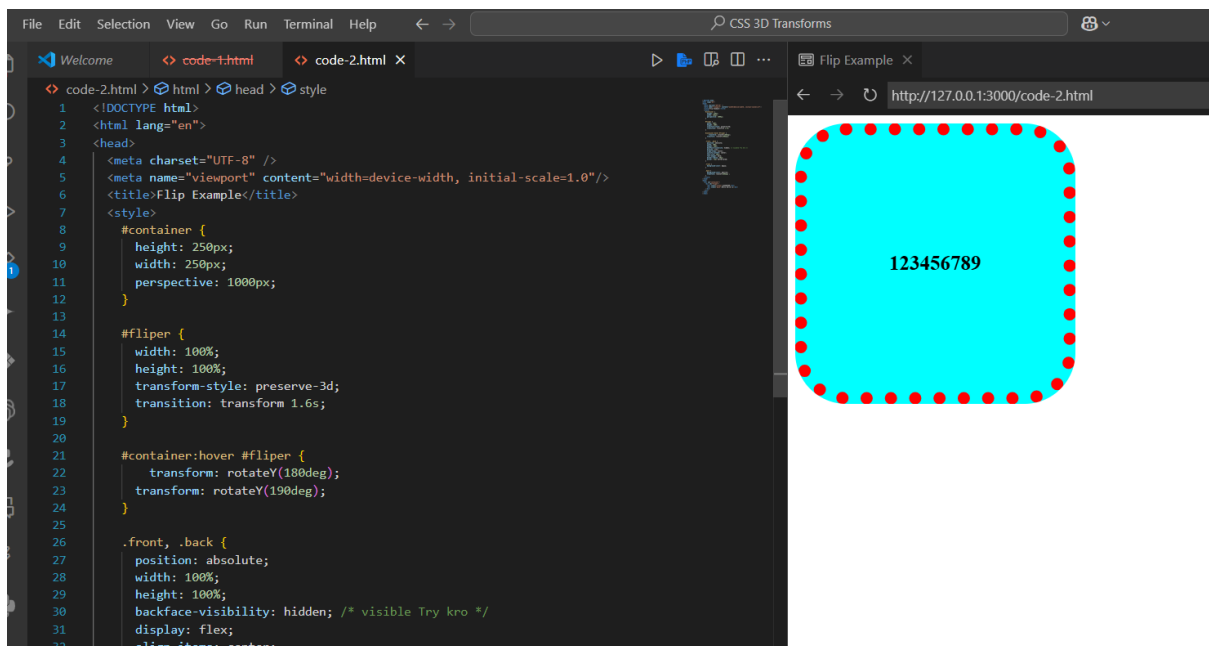
perspective: 300px = zyada zoom
perspective: 1000px = halka zoom

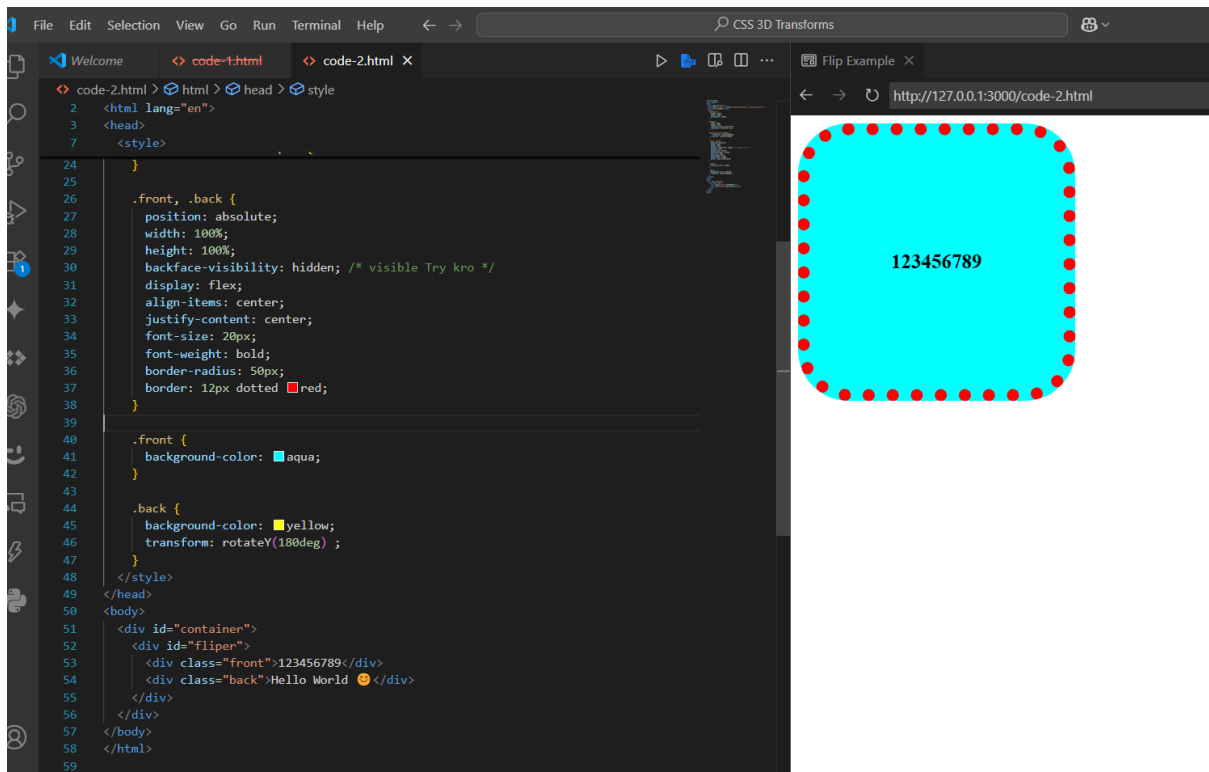
🔄 Example: 3D Flip Card

Click me ➡

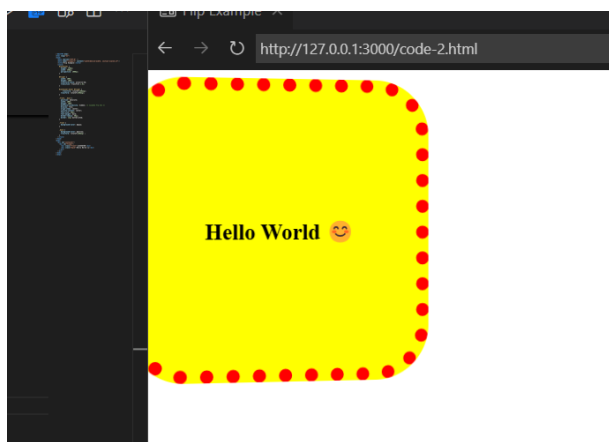


Code -2





After hover



Click me

```
<style>
.flip-container {
  perspective: 1000px;
  width: 200px;
  height: 200px;
}
```

```

.flipper {
  width: 100%;
  height: 100%;
  position: relative;
  transition: 0.6s;
  transform-style: preserve-3d;
}

.flip-container:hover .flipper {
  transform: rotateY(180deg);
}

.front, .back {
  position: absolute;
  width: 100%;
  height: 100%;
  backface-visibility: hidden;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 18px;
  color: white;
}

.front {
  background: #3498db;
}

.back {
  background: #e74c3c;
  transform: rotateY(180deg);
}
</style>

<div class="flip-container">
  <div class="flipper">
    <div class="front">👤 Front Side</div>
    <div class="back">👤 Back Side</div>
  </div>
</div>

```

Summary:

- rotateX() → X-axis (up/down palti)
 - rotateY() → Y-axis (left/right ghuma)
 - rotateZ() → Normal 2D rotate
 - perspective: → Parent pe lagao for depth
 - transform-style: preserve-3d; → Childs ko bhi 3D dikhne do
 - backface-visibility: hidden; → Piche wala invisible banao
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